Ansh Chandnani

ansh.chandnani@gmail.com | LinkedIn | github.com/mrdebator | Blog | anshc.me

Work Experience

Google **Sunnyvale, CA**

- *Security Engineer, Offensive Security* *July 2023 - Present*
  - Strengthen the security posture of core enterprise products by identifying **vulnerabilities** and driving remediation efforts through the design and execution of novel **red team exercises**.
  - Increase red team operator effectiveness by architecting **custom tooling** for the **reconnaissance** and **post-exploitation** phases of penetration tests.
  - Drive **cross-functional security initiatives**, including design reviews, vulnerability taxonomy development, and **strategic remediation planning.**
  - Mentor and guide an intern in validating **Windows Tactics, Techniques, and Procedures (TTPs)**, leading to improved **Active Directory** hygiene.
- *Security Engineering Intern, Offensive Security* *June 2022 - September 2022*
  - Authored a comprehensive **knowledge base** on Google's corporate and production environments by synthesizing **Open Source Intelligence (OSINT)**, enhancing the realism of red team operations.
  - Developed an automated OSINT pipeline and query-capable knowledge graph to analyze **1.4M+ systems**, enabling rapid identification of high-value targets and potential foothold vectors into **Google's production environment.**
  - Identified **insider risks** to critical production infrastructure through a specialized penetration test, leading to a **broader initiative** for stronger **credential hygiene.**
- *Security Engineering Intern, Detection Team* *June 2021 - September 2021*
  - Designed and built an **on-demand, platform-agnostic** validation tool (Go, Docker, GCP) to **standardize testing** for Zeek-based threat detection automation, slashing test cycle to **under 60 seconds.**
  - Contributed to corporate security by performing **threat detection** and **log analysis** for a task force focused on mitigating **insider threats** and policy violations.

Susquehanna International Group **Bala Cynwyd, PA**

- *Cybersecurity Coop, Firewall Services* *September 2020 - March 2021*
  - **Automated firewall rule lookups** by architecting a Python-based tool with custom APIs, significantly increasing operational efficiency for network security teams.
  - Managed and configured firewall rules across a global network of **offices and high-frequency trading exchanges**, ensuring secure and compliant network access for critical services.
  - Contributed to automation projects for firewall health monitoring and status alerting, directly supporting operational **uptime** of network infrastructure during **trading hours.**

Publications & Research

Authorships

⇒ " Malware Detection in Cloud Native Environments ", **ACM AICCC 2024**

⇒ " Algorithms: Decoding Bias in Messages, *in Media Literacy for Justice* ", **American Library Association**

⇒ " Ontology Modeling of Industrial Control System Ethical Hacking ", **ICCWS 2021**

Acknowledgements

⇒ Acknowledged for foundational contributions and primary research for the **Google DeepMind** paper, "A Framework for Evaluating Emerging Cyberattack Capabilities of AI "

Community & Leadership

- **Co-organizer & Judge**, BSides San Francisco CTF - Contributed to a premier global competition for **1,350+ participants** by designing web exploitation, forensics, and cryptography challenges and evaluating contestant submissions.
- Featured Speaker, **HackTheBox CubeCast Podcast** - Chronicled my journey in cybersecurity and red teaming during an interactive live stream, fielding real-time Q&A from 100+ attendees before the episode's release on Spotify.
- **Alumni Reviewer, Undergraduate Research & Enrichment Programs**, Drexel University - Evaluated and scored technical, undergraduate STEM research applications for Drexel's STAR Scholars program.
- **Judge**, **Startup in a Weekend Hackathon** organized by The Foundry - Evaluated technical demonstrations and implementations for 40+ teams during a high-stakes, 36-hour buildathon with over 120 participants.
- **President** (2022-23) & **Vice-President** (2021-22), Drexel Cyberdragons - Led **Drexel's cybersecurity club** for two years and architected a custom-sandboxed lab environment that provided hands-on training for 500+ students.
- **1st Place**, Central Regional Collegiate Penetration Testing Competition (CPTC) 2021
- Top 4 of 113 teams, CyberSEED Capture-the-Flag Competition 2021

Talks & Features

- eBPF & AI Powered EDRs: Can Red Teams Still Win? - Speaker, **Red Team Summit 2025**
- Cybersecurity 101 - **Guest Lecturer** , Northeastern University
- AI in Project Management - **Guest Lecturer**, Drexel University
- What is it really like to intern at a Big Tech company? - Featured in **Technical.ly Philly**, 2023
- FAANG Internship Panel - Invited Panelist, **Society of Asian Scientists and Engineers (SASE)**
- Cracking the Security Engineering Interview - Featured Security Engineer, **Exponent** (YouTube)

Select Projects

Claros *Python, Ontology/RDF, SPARQL*

Architected a **query-driven knowledge graph** and **OSINT automation** engine to empower security analysis, constructing a semantic graph to reveal complex attack paths between entities.

Outcome: Awarded "**Best Cybersecurity Project**" and "Second-runner up" at Drexel University's Senior Design showcase.

sysmonitor-ebpf *C, Go, eBPF*

Co-designed a high-performance, **kernel-level** system monitor for **cloud-native** environments using eBPF to provide holistic, namespace-aware visibility into both system calls and network events with extremely low overhead.

Outcome: This research served as the foundation for a peer-reviewed paper published at **ACM AICCC 2024**.

Cybersecurity Training Lab *Proxmox, Linux, UniFi Networking, Tailscale*

As President of Drexel's cybersecurity club, architected a **sandboxed lab** with vulnerable machines and attacker VMs on recycled enterprise servers; accessible via a dedicated local Wi-Fi network for easy, VPN-free access.

Outcome: Provided a safe, hands-on learning environment for **500+ students**.

Education

Drexel University June 2023

B.S. in Computing & Security Technology, minor in Computer Science GPA: 3.91/4.0

Honors:

- Summa Cum Laude
- Distinguished Service Award for Outstanding Leadership, 2023
- A* Award, 2023
- Best Poster Presentation, Stanford Research Conference, 2021
- Presented Research, Harvard National Collegiate Research Conference, 2021

Certifications & Trainings

- Fuzzing the Linux Kernel, RingZer0
- Exploiting the Linux Kernel, RingZer0
- Creative Red Teaming, Mandiant
- OSINT Corporate Recon, HackTheBox Academy
- Security Fundamdentals, Microsoft Training Association